10. The mathematically inclined reader will note that light travels along null geodesics of the spacetime metric, which, for definiteness, we can take to be $ds^2 = dt^2 - a^2(t)(dx^2)$, where $dx^2 = dx_1^2 + dx_2^2 + dx_3^2$, and the x_i are comoving coordinates. Setting $ds^2 = 0$, as appropriate for a null geodesic, we can write $\int_t^{t0} (dt/a(t))$ for the total comoving distance light emitted at time t can travel by time t_0. If we multiply this by the value of scale factor $a(t_0)$ at time t_0, then we will have calculated the physical distance that the light has traveled in this time interval. This algorithm can be widely used to calculate how far light can travel in any given time interval, revealing whether two points in space, for example, are in causal contact. As you can see, for accelerated expansion, even for arbitrarily large t_0, the integral is bounded, showing that the light will never reach arbitrarily distant comoving locations. Thus, in a universe with accelerated expansion, there are locations with which we can never communicate, and conversely, regions that can never communicate with us. Such regions are said to be beyond our cosmic horizon.

11. When analyzing geometrical shapes, mathematicians and physicists use a quantitative approach to curvature developed in the nineteenth century, which today is part of a mathematical body of knowledge known as differential geometry. One nontechnical way of thinking about this measure of curvature is to study triangles drawn on or within the shape of interest. If the triangle's angles add up to 180 degrees, as they do when it is drawn on a flat tabletop, we say the shape is flat. But if the angles add up to more or less than 180 degrees, as they do when the triangle is drawn on the surface of a sphere (the outward bloating of a sphere causes the sum of the angles to exceed 180 degrees) or the surface of a saddle (the inward shrinking of a saddle's shape causes the sum of the angles to be less than 180 degrees), we say the shape is curved. This is illustrated in Figure 8.6.

12. If you were to glue the opposite vertical edges of a torus together (which is reasonable to do, since they are identified—when you pass through one edge you immediately reappear on the other) you'd get a cylinder. And then, if you did the same for the upper and lower edges (which would now be in the shape of circles), you'd get a doughnut. Thus, a doughnut is another way of thinking about or representing a